Lattice Yang–Mills theory at strong coupling IV: the mass gap is $4\log(1/\beta) + O(1)$

A sequel to the foundations monograph of the project "Solving Yang–Mills and the Mass Gap"

9 November 2026; revised after an independent referee report, and the matching upper bound added (Section 5), the same day; living document

Abstract

For lattice Yang–Mills theory with Wilson's action in dimension $d \geq 3$ and $0 < \beta < \beta_0(d)$ (the threshold of the cluster expansion of [F]), we prove that the mass gap of the infinite-volume theory of Part II, and the finite-volume gaps of [F] on spatial lattices with all sides ≥ 4, are at least $4\log(1/q(\beta))$, with $q(\beta) = e^3 K(e^{4\beta} - 1)$ and $K = 4(2d-3)$. As $\beta \downarrow 0$ this is $4\log(1/\beta) - O(1)$. The leading coefficient 4 is the one predicted by the strong-coupling expansion (the lightest glueball is a tube with four plaquettes per unit of time, with predicted mass $-4\log u(\beta) + o(1)$, $u(\beta) \sim \beta/4$ for $SU(2)$), whereas the Dobrushin bounds of [F] and Part II give only $\log(1/\beta) - O(1)$. The proof combines three ingredients: duplicate variables, by which every term of the expansion of a covariance that does not connect the two observables cancels by the swap symmetry; the loop expansion and the ratio bound of Part III for the duplicated system; and a geometric lemma: a surface without free edges has at least four temporal plaquettes in each time layer it crosses, because these plaquettes form a graph of minimum degree two on the spatial lattice, whose girth is four (sides ≥ 4 are needed: with a side of length 3 the girth is three). In $d = 2$ the girth is the spatial size, which gives an infinite mass gap in infinite volume and a finite-volume gap growing linearly with the spatial size, in agreement with the exact solution. As in [F], II and III nothing non-abelian is used, so the result belongs to the strong-coupling endpoint of the problem. We also prove the matching upper bound: for an irreducible ρ of dimension N, $m \leq 4\log(1/u(\beta)) + O(\beta)$ with $u = c_\rho/(Nc_0)$, by an exact evaluation of the leading term of a plaquette–plaquette correlation (the four sides of a unit cube), a proof that every other term of the duplicated expansion has at least five plaquettes, and the spectral inequality $\mathrm{Cov} \leq e^{-m}\mathrm{Var}$ of the ground-state chain of Part II. Together: $m(\mu_\beta) = 4\log(1/\beta) + O(1)$ as $\beta \downarrow 0$, which is what principle 3 ("the strong-coupling test") of the research program requires. The upper bound is proved only for extremely small β (of order 10^{-16} for $SU(2)$ in $d = 4$), an artefact of sup-norm counting; its content is the asymptotics.

Contents

1 Introduction and main result **2**

2 The duplicated system **3**

3 Geometry: four plaquettes per time layer **5**

4 The covariance bound and the mass gap **6**

5 The matching upper bound **8**

6 Two dimensions; discussion 11

1 Introduction and main result

The foundations monograph [F] proves that the finite-volume mass gap is bounded below uniformly in the volume for small β, by Dobrushin's method; Part II [II] constructs the infinite-volume theory, identifies its transfer operator with a reversible Markov chain of time slices, and proves the same kind of bound, $m \geq \log(1/(6(d-1)\tanh\beta))$, for the infinite-volume mass gap $m(\mu_\beta)$. These bounds behave like $\log(1/\beta)$ as $\beta \downarrow 0$. The strong-coupling expansion predicts $m(\beta) = -4\log u(\beta) + O(u)$, where $u(\beta) = c_\rho(\beta)/(Nc_0(\beta))$ ($\approx \beta/4$ for $SU(2)$, $\approx \beta/(2N^2)$ for $N \geq 3$): the lightest state in the gauge-invariant sector propagates as a tube of plaquettes with four plaquettes per unit of Euclidean time. Principle 3 of the research program (`docs/03-research-program.md`, §3.1) asks every argument in the strong-coupling regime to reproduce this behaviour. Here we prove a lower bound with the correct leading coefficient for all $\beta < \beta_0(d)$ (Theorem 1.2) and, for sufficiently small β, the matching upper bound (Theorem 1.3); together, $m(\mu_\beta) = 4\log(1/\beta) + O(1)$.

Literature. The strong-coupling glueball spectrum was studied rigorously in the 1980s: by their abstracts, Schor [Sch83] proved the existence of glueballs in strongly coupled lattice gauge theories, and O'Carroll, Braga and Schor [OBS85] (in $2+1$ dimensions) and O'Carroll and Barbosa [OB85] (in $3+1$ dimensions) obtained isolated masses of the form $-4\log\beta + r(\beta)$. We have not yet read these papers (they are marked "verify"), and nothing below relies on them. What this article adds for the project is a short self-contained proof with explicit constants and range, for the infinite-volume theory of [II], and a covariance bound for all bounded local observables.

Convention 1.1 (Labels). As in [F, Convention 'Labels']: every Theorem, Proposition, Lemma and Corollary below has a complete proof here or in [F, II, III]; Remarks are commentary. We use the notation of [III]: $K := 4(2d-3)$, $\epsilon(\beta) = e^{2\beta} - 1$, $x(\beta) = e^2 K\epsilon(\beta)$, $\theta(\beta) = x/(1-x)$, and $\beta_0(d) = \frac{1}{2}\log(1 + \frac{1}{e^2K(K+2)})$, the threshold below which [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)] holds.

Theorem 1.2 (Main theorem). *Let G, ρ be as in [F, Convention 'Group'] with G metrisable, $d \geq 3$, $0 < \beta < \beta_0(d)$, and*

$$q(\beta) := e^3 K(e^{4\beta} - 1).$$

Then $q(\beta) < \frac{e(2+\epsilon_0)}{K+2} < 0.389$, where $\epsilon_0 = \frac{1}{e^2K(K+2)}$, and:

(i) for every periodic spatial lattice Λ_s with all sides $L_i \geq 4$, the finite-volume mass gap of [F, Definition 'Hamiltonian and mass gap' (`def:H`)] satisfies $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$;

(ii) the mass gap of the infinite-volume theory [II, Definition 'Mass gap of an infinite-volume lattice theory' (`iv:def:gap`)] satisfies $m(\mu_\beta) \geq 4\log(1/q(\beta))$;

(iii) for bounded measurable A, B on $G^{E(\mathbb{Z}^d)}$ depending only on finite link sets S_A, S_B, all of whose endpoints have time coordinates in $[a_1, a_2]$, resp. $[b_1, b_2]$, with $a_2 < b_1$,

$$|\mathrm{Cov}_{\mu_\beta}(A, B)| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q(\beta)}\, c(\beta)^{|S_A|+|S_B|}\, q(\beta)^{4(b_1-a_2)},$$

where $c(\beta) := (e^{2\theta(\beta)}(1 + \frac{1}{K}))^{2(d-1)}$.

As $\beta \downarrow 0$, $4\log(1/q(\beta)) = 4\log(1/\beta) - 4\log(4e^3K) + O(\beta)$.

Theorem 1.3 (Upper bound; the gap is $4\log(1/\beta)+O(1)$)**.** *Let $d \geq 3$, let ρ be irreducible (of dimension N), put $\iota := 1$ if $\rho \cong \bar{\rho}$ and $\iota := 0$ otherwise, and $u(\beta) := c_\rho(\beta)/(Nc_0(\beta))$. There are explicit functions $\delta(\beta), D(\beta) = O(\beta)$ (Theorem 5.4) such that, whenever $0 < \beta < \beta_0(d)$ and $\delta(\beta) < e^{-48(d-1)\theta(\beta)}$,*

$$m(\mu_\beta) \leq 4\log\frac{1}{u(\beta)} + D(\beta),$$

and the same bound holds for $m_{\Lambda_s}(\beta)$ for every periodic spatial lattice with all sides ≥ 3. Consequently, as $\beta \downarrow 0$,

$$-4\log(4e^3K) + O(\beta) \;\leq\; m(\mu_\beta) - 4\log\frac{1}{\beta} \;\leq\; 4\log\frac{2N^2}{1+\iota} + O(\beta),$$

and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the spatial lattices with all sides ≥ 4.

Here $\mathrm{Cov}(A,B) := \mu(AB) - \mu(A)\mu(B)$. Statement (iii) is the infinite-volume form of the covariance bound of Theorem 4.1; (i) follows from Theorem 4.1 by [F, Corollary 'Gap from Euclidean correlations' (`cor:gapfromcorr`)], and (ii) from (iii) by [II, Theorem 'OS reconstruction of the lattice theory' (`iv:thm:OS`)](iii).

Remark 1.4 (Comparison). For $d = 4$: the bound of Part II, $\log(1/(18\tanh\beta))$, is $\log(1/\beta) - 2.89 + O(\beta^2)$; the bound here is $4\log(1/\beta) - 29.5 + O(\beta)$. In the common range of validity $\beta < \beta_0(4) \approx 1.54 \cdot 10^{-4}$ the two are comparable near $\beta_0(4)$ (5.890 and 5.591 at $\beta_0(4)$), and the present one is larger for $\beta < 1.39 \cdot 10^{-4}$, by an additive amount that tends to infinity as $\beta \to 0$ (the ratio of the two bounds increases to 4; it is 1.92 at $\beta = 10^{-5}$). These values were computed with `python3`. The point of the theorem is the coefficient 4, not the numerical strength at fixed β; the additive constant -29.5 is far from the predicted $+4\log 4$ for $SU(2)$ (from $u \approx \beta/4$).

Strategy. (1) For the product measure $\mu\otimes\mu$ of two independent copies (U, U'), $2\,\mathrm{Cov}_\mu(A,B) = \int (A - A')(B - B')\, d(\mu\otimes\mu)$. The duplicated Wilson weight is $\prod_p (1 + f_\beta(U_p))(1 + f_\beta(U'_p)) = \prod_p (1 + g_p)$ with $|g_p| \leq e^{4\beta} - 1$, so the loop expansion of [III] applies to the duplicated system, with partition functions $\Xi(\cdot)^2$. (2) In this expansion every set of plaquettes that does not contain a component connecting the supports of A and B gives a product of two integrals, each odd under the exchange $U \leftrightarrow U'$, hence zero (Lemma 2.5). (3) A connecting component has no free edges away from the supports, so in every time layer between the supports it contains at least four temporal plaquettes (Lemma 3.2), hence at least $4\times$(number of layers) plaquettes (Corollary 3.4). (4) The sum over such sets is controlled by the ratio bound and the rooted counting of [III]. (5) For the upper bound (Section 5) the same expansion is applied to two plaquettes at times 0 and 1: the four sides of the unit cube between them form the unique minimal connecting set, its contribution is computed exactly by character orthogonality, all other terms have at least five plaquettes, and the spectral inequality $\mathrm{Cov} \leq e^{-m}\,\mathrm{Var}$ converts the resulting lower bound on the covariance into an upper bound on m.

2 The duplicated system

Throughout this section and the next, $\Lambda = \prod_{\lambda=0}^{d-1} \mathbb{Z}/L_\lambda\mathbb{Z}$ is a periodic lattice with all $L_\lambda \geq 3$, with the conventions of [F, III]: $\mathcal{P} = \mathcal{P}(\Lambda)$, $E(\Lambda)$, $E(p)$, $E(X)$, adjacency and connectedness of sets of plaquettes, $f_\beta = w_\beta/c_0(\beta) - 1$, and for $\mathcal{Q} \subseteq \mathcal{P}$ the restricted partition function $\Xi(\mathcal{Q}) = \int \prod_{p\in\mathcal{Q}}(1 + f_\beta(U_p))\, dU$ of [III, Definition 'Restricted partition functions, loop polymers' (`al:def:Xi`)], so that $\Xi(\mathcal{P}) = Z_\Lambda(\beta)/c_0(\beta)^{|\mathcal{P}|}$.

Definition 2.1 (Duplicated system)**.** On $\Omega_\Lambda \times \Omega_\Lambda$ with the product Haar measure $dU\,dU'$ put, for $p \in \mathcal{P}$,

$$g_p(U,U') := \big(1 + f_\beta(U_p)\big)\big(1 + f_\beta(U'_p)\big) - 1.$$

For $X \subseteq \mathcal{P}$ and bounded measurable F on $\Omega_\Lambda \times \Omega_\Lambda$ let $\Phi_F(X) := \iint F \prod_{p\in X} g_p \, dU\, dU'$.

Lemma 2.2 (Properties of the duplicated system)**.** *(a)* $\|g_p\|_\infty \le \epsilon(2\beta) = e^{4\beta} - 1$.

(b) For $\mathcal{Q} \subseteq \mathcal{P}$, $\iint \prod_{p\in\mathcal{Q}}(1+g_p)\, dU\, dU' = \Xi(\mathcal{Q})^2$*; and* $\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}$ *has density* $\prod_{p\in\mathcal{P}}(1+g_p)/\Xi(\mathcal{P})^2$ *with respect to* $dU\,dU'$.

(c) Let $X \subseteq \mathcal{P}$*, let* ℓ *be a link lying in exactly one plaquette of* X*, and let* H *be bounded measurable and independent of* (U_ℓ, U'_ℓ)*. Then* $\iint H \prod_{p\in X} g_p\, dU\, dU' = 0$.

Proof. (a) By [III, Lemma `al:lem:facts`](a),(b), $e^{-\beta} \le w_\beta \le e^\beta$ and $1 \le c_0 \le e^\beta$, so $1+f_\beta = w_\beta/c_0 \in [e^{-2\beta}, e^\beta]$ and $(1+f_\beta(U_p))(1+f_\beta(U'_p)) \in [e^{-4\beta}, e^{2\beta}]$ and $g_p \in [e^{-4\beta}-1, e^{2\beta}-1]$; both $1-e^{-4\beta}$ and $e^{2\beta}-1$ are at most $e^{4\beta}-1$. (b) The integrand is $\prod_{p\in\mathcal{Q}}(1+f_\beta(U_p))\cdot\prod_{p\in\mathcal{Q}}(1+f_\beta(U'_p))$; integrate over U and U' separately. For the density: $W_\beta(U)W_\beta(U') = c_0^{2|\mathcal{P}|}\prod_p(1+g_p)$ and $Z_\Lambda^2 = c_0^{2|\mathcal{P}|}\Xi(\mathcal{P})^2$. (c) Let p be the plaquette of X containing ℓ. Then $U_p = AU_\ell^{\pm1}B$ and $U'_p = A'U'^{\pm1}_\ell B'$ with A, B, A', B' independent of (U_ℓ, U'_ℓ), and no other factor of the integrand depends on (U_ℓ, U'_ℓ). By Fubini it suffices that $\iint g_p\, dU_\ell\, dU'_\ell = \int(1 + f_\beta(AU^{\pm1}B))\, dU \cdot \int(1 + f_\beta(A'U^{\pm1}B'))\, dU - 1 = 1\cdot 1 - 1 = 0$, by [III, Lemma `al:lem:facts`](c). □

Definition 2.3 (Loop polymers for a set of links)**.** For $S \subseteq E(\Lambda)$ let $\mathcal{P}_S := \{p : E(p) \cap S \ne \emptyset\}$, let $\mathcal{X}_S$ be the family of sets $X \subseteq \mathcal{P}$ each of whose connected components contains a plaquette of $\mathcal{P}_S$ (including $X = \emptyset$), and for $X \subseteq \mathcal{P}$ let $\overline{X} := \{p : E(p) \cap (E(X) \cup S) \ne \emptyset\}$. Put $J_S := 2(d-1)|S|$. Then $|\mathcal{P}_S| \le J_S$ and $|\overline{X}| \le (K+1)|X| + J_S$ (each link lies in $2(d-1)$ plaquettes, and each plaquette is adjacent to at most K plaquettes, [III, Lemma `al:lem:facts`](d)).

Lemma 2.4 (Duplicated loop expansion)**.** *Let* $S \subseteq E(\Lambda)$ *and let* F *be bounded, measurable, and dependent only on the variables* (U_ℓ, U'_ℓ)*,* $\ell \in S$*. Then*

$$\int F\, d(\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}) = \sum_{X\in\mathcal{X}_S} \Phi_F(X)\Big(\frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}\Big)^2.$$

Proof. This is the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] with f_β replaced by g, single link variables by pairs (U_ℓ, U'_ℓ), dU by $dU\,dU'$, and E_C by S. By Lemma 2.2(b), $\Xi(\mathcal{P})^2 \int F\, d(\mu\otimes\mu) = \sum_{T\subseteq\mathcal{P}} \iint F \prod_{p\in T} g_p\, dU\, dU'$. For $T \subseteq \mathcal{P}$ let $X(T)$ be the union of the components of T meeting $\mathcal{P}_S$ and $Y(T) := T \setminus X(T)$. The proof of the claim in [III, Lemma `al:lem:loopexp`], which uses only the combinatorics of components, of $\mathcal{P}_S$ (there $\mathcal{P}_C$) and of $\overline{X}$, shows that $T \mapsto (X(T), Y(T))$ is a bijection onto the pairs with $X \in \mathcal{X}_S$ and $Y \subseteq \mathcal{P}\setminus\overline{X}$. For such a pair, $F\prod_{p\in X} g_p$ depends only on the variables of the links in $S \cup E(X)$ and $\prod_{p\in Y} g_p$ only on those in $E(Y)$, a disjoint set; so the integral factorises, and $\sum_{Y\subseteq\mathcal{P}\setminus\overline{X}} \iint \prod_{p\in Y} g_p = \iint \prod_{p\in\mathcal{P}\setminus\overline{X}}(1+g_p) = \Xi(\mathcal{P}\setminus\overline{X})^2$ by Lemma 2.2(b). Divide by $\Xi(\mathcal{P})^2$. □

Lemma 2.5 (Cancellation)**.** *Let* A, B *be bounded measurable functions on* Ω_Λ *depending only on disjoint link sets* S_A, S_B*, put* $S := S_A \cup S_B$ *and* $F(U,U') := (A(U)-A(U'))(B(U)-B(U'))$*. Then:*

(i) $\int F\, d(\mu\otimes\mu) = 2\,\mathrm{Cov}_\mu(A,B)$ *for every probability measure* μ *on* Ω_Λ*, and* $\|F\|_\infty \le 4\|A\|_\infty\|B\|_\infty$*;*

(ii) if $X \in \mathcal{X}_S$ and no connected component of X contains both a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$, then $\Phi_F(X) = 0$;

(iii) if some link of $E(X) \setminus S$ lies in exactly one plaquette of X, then $\Phi_F(X) = 0$.

Proof. (i) Expand: $\int (A - A')(B - B')\, d(\mu \otimes \mu) = 2\mu(AB) - 2\mu(A)\mu(B)$. (ii) Let X_1 be the union of the components of X that contain a plaquette of $\mathcal{P}_{S_A}$, and $X_2 := X \setminus X_1$; by hypothesis no component of X_1 meets $\mathcal{P}_{S_B}$, and every component of X_2 meets $\mathcal{P}_{S_B}$ but not $\mathcal{P}_{S_A}$. A component containing a link of S_B contains a plaquette of $\mathcal{P}_{S_B}$; hence $E(X_1) \cap S_B = \emptyset$, and likewise $E(X_2) \cap S_A = \emptyset$; and $E(X_1) \cap E(X_2) = \emptyset$, since distinct components share no link. So $S_A \cup E(X_1)$ and $S_B \cup E(X_2)$ are disjoint, and since $dU\, dU'$ is a product measure,

$$\Phi_F(X) = \iint (A - A') \prod_{p \in X_1} g_p \cdot \iint (B - B') \prod_{p \in X_2} g_p.$$

The exchange $(U, U') \mapsto (U', U)$ preserves $dU\, dU'$ and every g_p (which is symmetric in U_p, U'_p), and changes the sign of $A - A'$; so the first factor equals its negative and vanishes. (iii) Apply Lemma 2.2(c) with $H = F \prod_{p' \in X \setminus \{p\}} g_{p'}$, p the plaquette containing the link. □

3 Geometry: four plaquettes per time layer

Time coordinates are taken in $\mathbb{Z}/L_0\mathbb{Z}$. A link $(x, 0)$ is *temporal of layer* x_0 (it joins times x_0 and $x_0 + 1$); a plaquette $(x, 0i)$, $1 \le i \le d-1$, is *temporal of layer* x_0. For $Y \subseteq \mathcal{P}$ and a layer s, Y_s denotes the set of temporal plaquettes of layer s in Y, and $T(p)$ the set of time coordinates of the corners of p (one time for a spatial plaquette, two consecutive times for a temporal one). Let $\Lambda_s := \prod_{i=1}^{d-1} \mathbb{Z}/L_i\mathbb{Z}$ be the spatial torus and g the *girth* of its nearest-neighbour graph (the length of a shortest cycle).

Lemma 3.1 (Girth)**.** *Let all $L_i \ge 3$. If $d \ge 3$, then $g = 4$ when all $L_i \ge 4$, and $g = 3$ when some $L_i = 3$. If $d = 2$, then $g = L_1$.*

Proof. Since all $L_i \ge 3$ the graph is simple. A cycle is a closed walk without backtracking, i.e. a sum of unit vectors $\pm e_i$ equal to 0 in Λ_s with no two consecutive steps opposite. For $d = 2$ the graph is the cycle $\mathbb{Z}/L_1\mathbb{Z}$ itself. For $d \ge 3$ a unit square is a cycle of length 4; a cycle of length 3 needs $\pm e_i \pm e_j \pm e_k = 0$ with no two opposite consecutive steps, which forces $i = j = k$ and $3e_i = 0$, i.e. $L_i = 3$; and conversely $L_i = 3$ gives the cycle $0, e_i, 2e_i$. □

Lemma 3.2 (Plaquettes per layer)**.** *Let all $L_i \ge 3$ $(1 \le i \le d-1)$. Let $S \subseteq E(\Lambda)$ and $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus S$ lies in at least two plaquettes of X. Let s be a layer none of whose temporal links belongs to S, and let Y be a connected component of X with $Y_s \neq \emptyset$. Then $|Y_s| \ge g$. In particular $|Y_s| \ge 4$ if $d \ge 3$ and all $L_i \ge 4$.*

Proof. The temporal links of layer s are $\ell_{\vec{x}} := ((s, \vec{x}), 0)$, $\vec{x} \in \Lambda_s$, and the plaquettes containing $\ell_{\vec{x}}$ are the temporal plaquettes $((s, \vec{x}), 0i)$ and $((s, \vec{x} - e_i), 0i)$ of layer s. Let Γ be the graph on Λ_s whose edges are the plaquettes $((s, \vec{x}), 0i) \in Y_s$, each joining $\vec{x}$ and $\vec{x} + e_i$ (the positions of its two temporal links). Since $L_i \ge 3$, distinct plaquettes give distinct edges and there are no loops, so Γ is a subgraph of the nearest-neighbour graph of Λ_s. If $\vec{x}$ is an endpoint of an edge, then $\ell_{\vec{x}} \in E(Y) \setminus S$, so $\ell_{\vec{x}}$ lies in at least two plaquettes of X; these are temporal plaquettes of layer s sharing the link $\ell_{\vec{x}}$ with a plaquette of Y, hence they belong to the component Y, i.e. to Y_s. So every non-isolated vertex of Γ has degree at least 2, and since $Y_s \neq \emptyset$, Γ contains a cycle (follow edges without immediately backtracking until a vertex repeats), which has at least g edges; they are distinct plaquettes of Y_s. The last statement is Lemma 3.1. □

Lemma 3.3 (Crossing layers)**.** *Let $Y \subseteq \mathcal{P}$ be connected and let $s \neq s'$ be layers with $Y_s = Y_{s'} = \emptyset$. Then either $T(p) \subseteq I_1$ for all $p \in Y$, or $T(p) \subseteq I_2$ for all $p \in Y$, where $I_1 := \{s+1, \dots, s'\}$ and $I_2 := \{s'+1, \dots, s\}$ are the two cyclic intervals of $\mathbb{Z}/L_0\mathbb{Z}$ determined by s, s'.*

Proof. I_1 and I_2 partition $\mathbb{Z}/L_0\mathbb{Z}$. A spatial plaquette has $T(p) = \{r\}$, contained in one of them. A temporal plaquette of layer $r \notin \{s, s'\}$ has $T(p) = \{r, r+1\}$, and $r, r+1$ lie in the same interval because the only consecutive pairs split between I_1 and I_2 are $\{s, s+1\}$ and $\{s', s'+1\}$. If $p, p' \in Y$ are adjacent, the time coordinates of the endpoints of a common link lie in $T(p) \cap T(p')$, so p, p' lie in the same class. By connectedness all plaquettes of Y lie in the same class. □

Corollary 3.4 (Size of connecting components)**.** *Let all $L_\lambda \geq 3$. Let $S_A, S_B \subseteq E(\Lambda)$ and integers $a_1 \leq a_2 < b_1 \leq b_2$ with $b_2 - a_1 < L_0$ be such that the endpoints of the links of S_A have time coordinates in $\{a_1, \dots, a_2\}$ and those of S_B in $\{b_1, \dots, b_2\}$ (modulo L_0). Put*

$$\tau_{\text{in}} := b_1 - a_2, \qquad \tau_{\text{out}} := L_0 - (b_2 - a_1), \qquad \tau := \min(\tau_{\text{in}}, \tau_{\text{out}}) \ \ (\geq 1).$$

Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus (S_A \cup S_B)$ lies in at least two plaquettes of X, and let Y be a component of X containing a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$. Then $|Y| \geq g\,\tau$. More precisely, with the inner layers $\{a_2, \dots, b_1 - 1\}$ and the outer layers $\{b_2, \dots, a_1 - 1 + L_0\}$, Y contains at least g temporal plaquettes in every inner layer, or at least g temporal plaquettes in every outer layer.

Proof. Call the layers $s \in \{a_2, \dots, b_1 - 1\}$ inner (τ_{in} of them) and the layers $s' \in \{b_2, \dots, a_1 - 1 + L_0\}$ outer (τ_{out} of them); as integers, $s < s' < s + L_0$ for every inner s and outer s', so inner and outer layers are distinct. *These layers are free.* A temporal link of an inner layer s joins s and $s+1$, where $s+1 \in \{a_2+1, \dots, b_1\}$ is not in $\{a_1, \dots, a_2\}$ and $s \in \{a_2, \dots, b_1 - 1\}$ is not in $\{b_1, \dots, b_2\}$ modulo L_0 (because $b_2 - a_1 < L_0$); so it belongs neither to S_A nor to S_B. The same argument applies to outer layers. *Times of the plaquettes of Y at the supports.* Let $p \in Y \cap \mathcal{P}_{S_A}$, containing a link $\ell \in S_A$. If p is spatial, $T(p) = \{r\}$ with $r \in \{a_1, \dots, a_2\}$; if p is temporal of layer r, then ℓ is either its temporal link of layer r or a spatial link at time r or $r+1$, so $r \in \{a_1 - 1, \dots, a_2\}$. Now let s be inner and s' outer with $Y_s = Y_{s'} = \emptyset$, and I_1, I_2 as in Lemma 3.3. Then $T(p) \subseteq I_2 = \{s'+1, \dots, s\}$: the times $a_1, \dots, a_2$ lie in I_2 because $s' + 1 \leq a_1 + L_0$ and $s \geq a_2$; the time $a_2 + 1$ occurs in $T(p)$ only if p is temporal of layer a_2, which is excluded when $s = a_2$ (as $Y_s = \emptyset$) and lies in I_2 when $s \geq a_2 + 1$; similarly the time $a_1 - 1$ occurs only if p is temporal of layer $a_1 - 1$, which is excluded when $s' \equiv a_1 - 1$ and lies in I_2 otherwise. In the same way $T(p) \subseteq I_1 = \{s+1, \dots, s'\}$ for every $p \in Y \cap \mathcal{P}_{S_B}$. *Conclusion.* If $Y_s = Y_{s'} = \emptyset$ for some inner s and some outer s', Lemma 3.3 puts all of Y into one of the disjoint classes I_1, I_2, which contradicts the previous paragraph, since Y meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Hence $Y_s \neq \emptyset$ for all inner layers, or $Y_{s'} \neq \emptyset$ for all outer layers. In either case Lemma 3.2 (with $S = S_A \cup S_B$) gives at least g plaquettes of Y in each of these layers, and plaquettes of different layers are different. □

4 The covariance bound and the mass gap

Theorem 4.1 (Covariance bound in finite volume)**.** *Let $d \geq 2$, $0 < \beta < \beta_0(d)$, and let Λ be a periodic lattice with all $L_\lambda \geq 3$; let g be the girth of Lemma 3.1. Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B as in Corollary 3.4, with $\tau = \min(\tau_{\text{in}}, \tau_{\text{out}})$. Then, with $q = q(\beta) < 1$ (Lemma 4.2) and $c = c(\beta)$ as in Theorem 1.2,*

$$\left|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, B)\right| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q}\, c^{|S_A| + |S_B|}\, q^{g\tau}.$$

Proof. If S_A or S_B is empty, A or B is constant and the covariance vanishes; so let both be non-empty. Let $S := S_A \cup S_B$ and F as in Lemma 2.5. By Lemmas 2.4 and 2.5, $2\operatorname{Cov}(A,B) = \sum_X \Phi_F(X)(\Xi(\mathcal{P}\setminus\overline{X})/\Xi(\mathcal{P}))^2$, where only sets $X \in \mathcal{X}_S$ contribute in which every link of $E(X)\setminus S$ lies in at least two plaquettes of X and some component meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$; by Corollary 3.4 these have $|X| \geq n_0 := g\tau$. For each of them: $|\Phi_F(X)| \leq 4\|A\|_\infty\|B\|_\infty\,\epsilon(2\beta)^{|X|}$ by Lemmas 2.5(i) and 2.2(a); and by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)] and Definition 2.3,

$$\Big(\frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}\Big)^2 \leq e^{2\theta|\overline{X}|} \leq e^{2\theta(K+1)|X|}e^{2\theta J_S} \leq e^{2|X|}e^{2\theta J_S},$$

using $(K+1)\theta < 1$ [III, Lemma `al:lem:constants`]. The number of $X \in \mathcal{X}_S$ with $|X| = n$ is at most $\binom{J_S+Kn}{n}$: this is [III, Corollary 'Counting loop polymers' (`al:cor:countloop`)] with E_C replaced by S (its proof uses only $|\mathcal{P}_S| \leq J_S$ and the adjacency bound K; it requires $J_S \geq 1$, i.e. $S \neq \emptyset$). With $\binom{m}{n}t^n \leq (1+t)^m$ at $t = 1/K$, $\binom{J_S+Kn}{n} \leq (1+\frac{1}{K})^{J_S}(1+\frac{1}{K})^{Kn}K^n \leq (1+\frac{1}{K})^{J_S}(eK)^n$. Altogether

$$2|\operatorname{Cov}(A,B)| \leq 4\|A\|_\infty\|B\|_\infty\, e^{2\theta J_S}\Big(1+\frac{1}{K}\Big)^{J_S}\sum_{n\geq n_0}\big(eK\cdot e^2\epsilon(2\beta)\big)^n = \frac{4\|A\|_\infty\|B\|_\infty\, c^{|S|}}{1-q}\,q^{n_0},$$

since $e^3K\epsilon(2\beta) = q$ and $(e^{2\theta}(1+\frac{1}{K}))^{J_S} = c^{|S|}$. □

Lemma 4.2 (Constants)**.** *For $d \geq 2$ and $0 < \beta < \beta_0(d)$: $q(\beta) < \frac{e(2+\epsilon_0)}{K+2}$, which is < 0.909 for $d = 2$ and < 0.389 for $d \geq 3$; and $4\log\frac{1}{q(\beta)} = 4\log\frac{1}{\beta} - 4\log(4e^3K) + O(\beta)$ as $\beta \downarrow 0$. Moreover $\beta_0(d) < \beta_*(d) = \operatorname{artanh}\frac{1}{6(d-1)}$, the threshold of [II].*

Proof. With $\epsilon = e^{2\beta}-1 < \epsilon_0$ (which is the condition $\beta < \beta_0(d)$), $e^{4\beta}-1 = \epsilon(2+\epsilon) < \epsilon_0(2+\epsilon_0)$ and $e^3K\epsilon_0 = \frac{e}{K+2}$; for $d = 2$, $K = 4$ and $\frac{e(2+\epsilon_0)}{6} < 0.909$; for $d \geq 3$, $K \geq 12$ and $\frac{e(2+\epsilon_0)}{14} < 0.389$. The asymptotics follow from $e^{4\beta}-1 = 4\beta+O(\beta^2)$. The last claim is [II, Corollary 'Area law in infinite volume' (`iv:cor:arealaw`)] (first line of its proof). □

Proof of Theorem 1.2. The bound on q is Lemma 4.2. (iii) Since $\beta < \beta_*(d)$, the infinite-volume Gibbs measure μ_β of [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)] exists and is the limit of the periodised measures $\tilde{\mu}_{\Lambda_k,\beta}$ on every bounded measurable local function, for every sequence of periodic lattices with all sides tending to infinity. Take such Λ_k with spatial sides ≥ 4. For k large, the reduction map is injective on the links of $S_A \cup S_B$, the functions $A\circ\iota$, $B\circ\iota$ on Ω_{Λ_k} (in the notation of [II]) depend on the images of S_A, S_B, which satisfy the hypotheses of Theorem 4.1 with the same a_i, b_i, $g = 4$, $\tau_{\text{in}} = b_1 - a_2$ and $\tau_{\text{out}} \to \infty$. Theorem 4.1 bounds $\operatorname{Cov}_{\tilde{\mu}_{\Lambda_k,\beta}}(A,B)$ uniformly in k, and the left side converges to $\operatorname{Cov}_{\mu_\beta}(A,B)$.

(ii) Let $F \in \mathcal{A}_+$ in the notation of [II, Definition 'Reflection, shifts, half-space algebras' (`iv:def:reflection`)], depending only on a finite set of links whose endpoints have time coordinates in $[0, n_F]$. Then ΘF depends only on links with endpoint times in $[-n_F, 0]$ and S_tF on links with endpoint times in $[t, t+n_F]$, with the same number of links. For $t \geq 1$ the two sets are disjoint and (iii) gives, by θ- and translation invariance of μ_β (so that $\mu_\beta(\Theta F)\mu_\beta(S_tF) = |\mu_\beta(F)|^2$),

$$\big|\mu_\beta(\Theta F\cdot S_tF) - |\mu_\beta(F)|^2\big| = \big|\operatorname{Cov}_{\mu_\beta}(\Theta F, S_tF)\big| \leq C_F\, q^{4t} \qquad (t \geq 1),$$

with C_F independent of t (here $b_1 - a_2 = t$); for $t = 0$ the left side is at most $2\|F\|_\infty^2$. So the hypothesis of [II, Theorem `iv:thm:OS`](iii) holds with $m = 4\log(1/q)$ and the constant $\max(C_F, 2\|F\|_\infty^2)$, and $m(\mu_\beta) \geq 4\log(1/q)$.

(i) Let A be a bounded gauge-invariant function on $\Omega_s = G^{E(\Lambda_s)}$ and $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ with $L_0 \geq 3$. The functions $\overline{A}(t)$ and $A(0)$ depend on the spatial links of the slices t and 0: in Corollary 3.4 take $a_1 = a_2 = 0$, $b_1 = b_2 = t$; for $1 \leq t \leq L_0/2$, $\tau = \min(t, L_0 - t) = t$, and $g = 4$ since all $L_i \geq 4$. Theorem 4.1 gives $|\langle \overline{A}(t)A(0)\rangle - \langle \overline{A}(t)\rangle\langle A(0)\rangle| \leq C_A q^{4t}$ with C_A depending on $\|A\|_\infty$ and $|E(\Lambda_s)|$ only, not on L_0; for $t = 0$ the left side is at most $2\|A\|_\infty^2$. [F, Corollary `cor:gapfromcorr`] (whose proof lets $L_0 \to \infty$ at fixed t, so that the bound is needed only for large L_0) gives $m_{\Lambda_s}(\beta) \geq 4\log(1/q)$. □

5 The matching upper bound

Throughout this section $d \geq 3$ and, in addition to the standing assumptions, ρ is irreducible; $N = d_\rho$, $\iota := 1$ if $\rho \cong \bar\rho$ and $\iota := 0$ otherwise, and $u(\beta) := c_\rho(\beta)/(Nc_0(\beta))$. Fix two spatial directions, labelled 1, 2. On a periodic lattice Λ (all $L_\lambda \geq 3$) let $p_0 := (0, 12)$ be the spatial plaquette at the origin and time 0, $p_1 := (e_0, 12)$ its translate at time 1, and

$$A := \operatorname{Re}\chi_\rho(U_{p_0}), \qquad A_1 := \operatorname{Re}\chi_\rho(U_{p_1}), \qquad S_A := E(p_0), \quad S_B := E(p_1), \quad S := S_A \cup S_B,$$

$F(U, U') := (A(U) - A(U'))(A_1(U) - A_1(U'))$ as in Lemma 2.5, and let X_0 be the set of the four temporal plaquettes $(y, 0i)$ of layer 0 with $(y, i) \in E(p_0)$: the sides of the unit cube Q with bottom p_0 and top p_1. The same notation is used on $\mathbb{Z}^d$.

Lemma 5.1 (Leading terms)**.** *For every $\beta > 0$:*

(i) $\tau_0 := \int A^2\, dU = \frac{1+\iota}{2}$;

(ii) $\tau_1 := \int A\, A_1 \prod_{s\in X_0} f_\beta(U_s)\, dU = \tau_0\, u(\beta)^4$;

(iii) $\Phi_F(X_0) = 2\tau_1$;

(iv) $\frac{(1+\iota)\beta e^{-\beta}}{2N^2} \leq u(\beta) \leq \frac{(1+\iota)\beta+\beta^2 e^\beta}{2N^2}$.

Proof. (i) With $\operatorname{Re}\chi = \frac12(\chi + \bar\chi)$, $\int(\operatorname{Re}\chi_\rho)^2 = \frac14(\int\chi_\rho^2 + 2\int|\chi_\rho|^2 + \int\overline{\chi_\rho}^2)$. By Schur orthogonality [F, Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`)](ii), $\int|\chi_\rho|^2 = 1$ and, since $\chi_{\bar\rho} = \overline{\chi_\rho}$, $\int\chi_\rho^2 = \int\chi_\rho\overline{\chi_{\bar\rho}} = \iota$; likewise $\int\overline{\chi_\rho}^2 = \iota$.

(ii) Orient the six faces of Q as its boundary. The computation in the proof of [F, Proposition 'Cube activity' (`prop:cube`)] shows: for irreducible representations attached to the six faces, $\int\prod_{\text{faces}}\chi_{r_{\text{face}}}(U_{\text{face}})\, dU$ equals d_r^{-4} if all six are the same r, and 0 otherwise. The functions f_β and $\operatorname{Re}\chi_\rho$ are invariant under inversion and conjugation, so A, A_1 and $f_\beta(U_s)$ are the same functions of the boundary-oriented holonomies. Insert $f_\beta = \sum_{r\neq0}(c_r/c_0)\chi_r$ (absolutely and uniformly convergent, [F, Corollary `cor:uniform`]) for the four sides, and $A = \chi_\rho$ if $\iota = 1$ (χ_ρ is then real), $A = \frac12(\chi_\rho + \chi_{\bar\rho})$ if $\iota = 0$, and the same for A_1. Only the terms in which all six faces carry the same $r \in \{\rho, \bar\rho\}$ survive. As $c_{\bar\rho} = \overline{c_\rho} = c_\rho$ ([F, Lemma 'Positivity of the character coefficients' (`lem:cr`)]) and $d_{\bar\rho} = d_\rho = N$, this gives $\tau_1 = (c_\rho/c_0)^4N^{-4} = u^4$ if $\iota = 1$ and $\tau_1 = \frac14\cdot2\cdot(c_\rho/c_0)^4N^{-4} = \frac12u^4$ if $\iota = 0$; in both cases $\tau_1 = \tau_0u^4$.

(iii) Expand $\prod_{s\in X_0} g_s = \prod_s(f_s + f_s' + f_sf_s')$, where $f_s := f_\beta(U_s)$ and $f_s' := f_\beta(U_s')$, and for each of the 3^4 terms let $Q_1 \subseteq X_0$ ($Q_2 \subseteq X_0$) be the set of sides carrying a factor f_s (a factor f_s'); $Q_1 \cup Q_2 = X_0$. Each of the four temporal links of Q lies in exactly two sides, adjacent around the square, and in no other factor of the integrand. If some temporal link lies in exactly one plaquette of Q_1, integrating it out in the first copy gives 0 ([III, Lemma `al:lem:facts`](c)); so $Q_1 \in \{\emptyset, X_0\}$, and likewise $Q_2 \in \{\emptyset, X_0\}$. Three terms remain: $\prod f_s$, $\prod f_s'$ and $\prod f_sf_s'$. Expand $F = AA_1 - AA_1' - A'A_1 + A'A_1'$; the integral over $dU\, dU'$ factorises into the two copies. A factor $\int A\prod_s f_s\, dU$, $\int A_1\prod_s f_s\, dU$ or $\int\prod_s f_s\, dU$ vanishes, because the top (resp. bottom) edges of Q then lie in exactly one factor; the same holds in the second copy.

Hence the term $\prod f_s$ contributes $\int AA_1 \prod_s f_s\, dU = \tau_1$, the term $\prod f'_s$ contributes τ_1, and the term $\prod f_s f'_s$ contributes 0.

(iv) By the proof of [F, Lemma `lem:cr`], $c_\rho = \sum_{n,m\ge 0} \frac{t^{n+m}}{n!\,m!} m_\rho(n,m)$ with $t = \beta/(2N)$ and $m_\rho(n,m)$ the multiplicity of ρ in $\rho^{\otimes n} \otimes \bar\rho^{\otimes m}$. Here $m_\rho(0,0) = 0$, $m_\rho(1,0) = 1$, $m_\rho(0,1) = \iota$, and $m_\rho(n,m) \le N^{n+m-1}$ (dimension count). Hence $(1+\iota)t \le c_\rho \le (1+\iota)t + N^{-1}(e^{2Nt} - 1 - 2Nt) \le \frac{(1+\iota)\beta}{2N} + \frac{\beta^2 e^\beta}{2N}$, using $e^y - 1 - y \le \frac{y^2}{2} e^y$ with $y = 2Nt = \beta$. Divide by Nc_0 and use $1 \le c_0 \le e^\beta$ ([III, Lemma `al:lem:facts`](b)). □

Lemma 5.2 (Minimal connecting set)**.** *Let Λ be periodic with all $L_\lambda \ge 3$. Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus S$ lies in at least two plaquettes of X, and let Y be a component of X containing a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$. Then $|Y| \ge 4$, with equality only if $Y = X_0$. Consequently, $\Phi_F(X) = 0$ for every $X \in \mathcal{X}_S$ with $|X| \le 4$ and $X \ne X_0$.*

Proof. Apply Corollary 3.4 with $a_1 = a_2 = 0$, $b_1 = b_2 = 1$: the only inner layer is 0, the outer layers are $1, \dots, L_0 - 1$ (at least two), and $g \ge 3$ (Lemma 3.1). If Y has temporal plaquettes in every outer layer, $|Y| \ge 3 \cdot 2 > 4$. Otherwise $|Y_0| \ge g \ge 3$. Suppose $|Y| \le 4$. The temporal plaquette $(y, 0i)$ of layer 0 contains exactly one spatial link at time 0, its bottom edge (y, i), and distinct such plaquettes have distinct bottom edges. A bottom edge of a plaquette of Y_0 lies in no plaquette of another component; it is not in S_B (whose links are at time 1); so, unless it lies in S_A, it lies in a second plaquette of Y, not in Y_0.

If $|Y_0| = 4$, then $Y = Y_0$, so all four bottom edges lie in $S_A = E(p_0)$; being four distinct links, they are the four edges of p_0, and $Y = X_0$.

If $|Y_0| = 3$, the graph Γ of the proof of Lemma 3.2 has three edges and minimum degree 2 at its non-isolated vertices, so it is a triangle; by Lemma 3.1 some $L_i = 3$ and the bottom edges are the three collinear links $(y, i), (y + e_i, i), (y + 2e_i, i)$ at time 0. At most one further plaquette p' belongs to Y, and p' contains at most one of them (the links of a plaquette in direction i are (z, i) and $(z + e_j, i)$ with $j \ne i$, and a temporal plaquette contains one spatial link at time 0). So at least two of the three lie in $S_A = E(p_0)$, which contains at most one link of a collinear triple (its links in direction $i \in \{1, 2\}$ are $(0, i)$ and (e_j, i), $\{i, j\} = \{1, 2\}$). This contradiction shows $|Y_0| \ne 3$.

For the last statement: if $\Phi_F(X) \ne 0$, Lemma 2.5 gives no free edges outside S and a component Y as above, so $|X| \ge |Y| \ge 4$, and $|X| = 4$ forces $X = Y = X_0$. □

Lemma 5.3 (Two-point bounds in finite volume)**.** *Let $0 < \beta < \beta_0(d)$ and let Λ be periodic with all $L_\lambda \ge 3$. Put*

$$\hat q(\beta) := eK\, e^{2(K+1)\theta(\beta)} \big(e^{4\beta} - 1\big) \;\le\; q(\beta) < 1, \qquad c(\beta) = \big(e^{2\theta(\beta)}(1 + \tfrac{1}{K})\big)^{2(d-1)}.$$

Then

$$\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, A_1) \ge \tau_1\, e^{-48(d-1)\theta} - \frac{2N^2 c^8 \hat q^{\,5}}{1 - \hat q},$$
$$\mathrm{Var}_{\mu_{\Lambda,\beta}}(A) \le \tau_0\, e^{8(d-1)\theta} + N^2 \big(e^\theta(1 + \tfrac{1}{K})\big)^{8(d-1)} \frac{x}{1 - x}.$$

Proof. Covariance. By Lemmas 2.4 and 2.5(i), $2\,\mathrm{Cov}(A, A_1) = \sum_{X \in \mathcal{X}_S} \Phi_F(X) \rho_X^2$ with $\rho_X := \Xi(\mathcal{P} \setminus \overline{X})/\Xi(\mathcal{P})$. The term X_0 equals $2\tau_1 \rho_{X_0}^2$ (Lemma 5.1(iii)), and $\rho_{X_0}^2 \ge e^{-2\theta|\overline{X_0}|}$ by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)]; since $S \subseteq E(X_0)$ and $E(X_0)$ consists of the twelve edges of Q, $\overline{X_0}$ is the set of plaquettes containing one of these edges, and $|\overline{X_0}| \le 24(d-1)$. By Lemma 5.2 every other non-zero term has $|X| \ge 5$. Bound them as in the proof of Theorem 4.1, but keeping the factor $e^{2\theta(K+1)|X|}$ instead of $e^{2|X|}$: with $J_S = 16(d-1)$ and $\|A\|_\infty, \|A_1\|_\infty \le N$,

$$\sum_{X \ne X_0} |\Phi_F(X)| \rho_X^2 \le 4N^2 e^{2\theta J_S} \Big(1 + \frac{1}{K}\Big)^{J_S} \sum_{n \ge 5} \big(eK\, e^{2\theta(K+1)} \epsilon(2\beta)\big)^n = \frac{4N^2 c^8 \hat q^{\,5}}{1 - \hat q}.$$

Divide by 2.

Variance. $\mathrm{Var}(A) \le \langle A^2\rangle$. The proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] applies verbatim to the function A^2, which depends only on the links of S_A, with E_C replaced by S_A; it gives $\langle A^2\rangle = \sum_{X\in\mathcal{X}_{S_A}} \Phi_{A^2}(X)\,\Xi(\mathcal{P}\setminus\overline{X})/\Xi(\mathcal{P})$ with $\Phi_{A^2}(X) := \int A^2 \prod_{p\in X} f_\beta(U_p)\,dU$. The term $X=\emptyset$ is $\tau_0\,\Xi(\mathcal{P}\setminus\mathcal{P}_{S_A})/\Xi(\mathcal{P}) \le \tau_0 e^{\theta|\mathcal{P}_{S_A}|} \le \tau_0 e^{8(d-1)\theta}$. For the others, $|\Phi_{A^2}(X)| \le N^2\epsilon^{|X|}$, the ratio is at most $e^{\theta(K+1)|X|+\theta J_{S_A}}$ with $J_{S_A} = 8(d-1)$, and there are at most $(1+\frac{1}{K})^{J_{S_A}}(eK)^n$ sets with $|X|=n$ (as in the proof of Theorem 4.1); since $eKe^{\theta(K+1)}\epsilon \le e^2K\epsilon = x$, their sum is at most $N^2(e^\theta(1+\frac{1}{K}))^{8(d-1)}\sum_{n\ge1}x^n$. □

Theorem 5.4 (Upper bound)**.** *Let $d\ge3$, ρ irreducible, $0<\beta<\beta_0(d)$, and with the notation of Lemma 5.3 suppose that*

$$\delta(\beta) := \frac{4N^2}{1+\iota}\,\frac{c(\beta)^8\,\hat q(\beta)^5}{(1-\hat q(\beta))\,u(\beta)^4} \;<\; e^{-48(d-1)\theta(\beta)}.$$

Put

$$D(\beta) := \log\frac{e^{8(d-1)\theta} + \frac{2N^2}{1+\iota}\big(e^\theta(1+\frac{1}{K})\big)^{8(d-1)}\frac{x}{1-x}}{e^{-48(d-1)\theta}-\delta(\beta)}.$$

Then

(i) $m(\mu_\beta) \le 4\log\frac{1}{u(\beta)} + D(\beta)$;

(ii) $m_{\Lambda_s}(\beta) \le 4\log\frac{1}{u(\beta)} + D(\beta)$ *for every periodic spatial lattice Λ_s with all sides ≥ 3.*

Moreover $\delta(\beta)=O(\beta)$ and $D(\beta)=O(\beta)$ as $\beta\downarrow0$; in particular the hypothesis holds for all sufficiently small β.

Proof. Write c_1 and c_0 for the covariance of A_1 and A and for the variance of A in the relevant theory. By Lemma 5.3, $\tau_0/\tau_1 = u^{-4}$ (Lemma 5.1) and $2N^2/\tau_1 = \frac{4N^2}{(1+\iota)u^4}$,

$$c_1 \ge \tau_1\big(e^{-48(d-1)\theta}-\delta(\beta)\big) > 0, \qquad c_0 \le \tau_0\Big(e^{8(d-1)\theta} + \frac{2N^2}{1+\iota}\big(e^\theta(1+\tfrac{1}{K})\big)^{8(d-1)}\frac{x}{1-x}\Big),$$

so $\log(c_0/c_1) \le 4\log(1/u) + D(\beta)$, provided these bounds hold for the theory in question. It remains to show $m \le \log(c_0/c_1)$.

(ii) Let $\Lambda = (\mathbb{Z}/L_0\mathbb{Z})\times\Lambda_s$ with $L_0\ge3$, and A the corresponding gauge-invariant function on the spatial configurations, so that $A_1 = A(1)$ in the notation of [F, Theorem 'Transfer matrix formalism' (`thm:tm`)]. By [F, Theorem 'Ground-state expectations' (`thm:gs`)], as $L_0\to\infty$, $\mathrm{Cov}_\Lambda(A_1,A)\to\langle Q\hat\Omega, e^{-H}Q\hat\Omega\rangle$ and $\mathrm{Var}_\Lambda(A)\to\|Q\hat\Omega\|^2$, where Q is the projection orthogonal to the vacuum, as in the proof of [F, Theorem 'Gap ⇔ clustering in time' (`thm:gapclust`)](i); and $\langle Q\hat\Omega, e^{-H}Q\hat\Omega\rangle \le e^{-m_{\Lambda_s}(\beta)}\|Q\hat\Omega\|^2$. The bounds of Lemma 5.3 are uniform in L_0, so they hold for the limits c_1, c_0, and $e^{-m_{\Lambda_s}(\beta)} \ge c_1/c_0$.

(i) By [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)](iii) (note $\beta<\beta_0(d)<\beta_*(d)$, Lemma 4.2), the covariance and the variance under μ_β are limits of those on periodic lattices with all sides tending to infinity, so the bounds of Lemma 5.3 hold for them. With ν and P as in [II, Corollary 'The time-slice chain' (`iv:cor:chain`)] and $\tilde A := A-\nu(A)$, a function of the time-zero slice, [II, Corollary `iv:cor:chain`](iv) gives $c_1 = \langle\tilde A, P\tilde A\rangle_\nu$ and $c_0 = \|\tilde A\|_\nu^2$; since $\tilde A\perp\mathbf{1}$ and $\|P|_{\mathbf{1}^\perp}\| = e^{-m(\mu_\beta)}$ [II, Theorem 'Transfer operator, ground-state chain and mass gap' (`iv:thm:gap`)], $c_1 \le e^{-m(\mu_\beta)}c_0$.

Finally, $c(\beta)$ is bounded, $\theta(\beta), x(\beta), \hat q(\beta) = O(\beta)$, and $u(\beta)\ge(1+\iota)\beta e^{-\beta}/(2N^2)$ (Lemma 5.1(iv)), so $\delta(\beta) = O(\hat q^5/\beta^4) = O(\beta)$ and $D(\beta)=O(\beta)$. □

Corollary 5.5 (The gap is $4\log(1/\beta)+O(1)$)**.** *Under the hypotheses of Theorem 5.4,*

$$4\log\frac{1}{\beta}-4\log(4e^3K)-4\log\frac{e^{4\beta}-1}{4\beta}\ \le\ m(\mu_\beta)\ \le\ 4\log\frac{1}{\beta}+4\log\frac{2N^2}{1+\iota}+4\beta+D(\beta),$$

and the same bounds hold for $m_{\Lambda_s}(\beta)$ for every periodic spatial lattice with all sides ≥ 4. In particular $m(\mu_\beta)=4\log(1/\beta)+O(1)$ as $\beta\downarrow 0$, and the same holds uniformly for the finite-volume gaps on spatial lattices with all sides ≥ 4.

Proof. The lower bounds are Theorem 1.2(ii) and (i), with $q(\beta)=4e^3K\beta\cdot\frac{e^{4\beta}-1}{4\beta}$. The upper bounds are Theorem 5.4 with $\log\frac{1}{u}\le\log\frac{2N^2}{(1+\iota)\beta}+\beta$ (Lemma 5.1(iv)). □

Remark 5.6 (The threshold of Theorem 5.4). Numerically (`python3`), $\delta(\beta)\approx 1.0\cdot 10^{16}\beta$ for $SU(2)$ in $d=4$, $\approx 1.0\cdot 10^{15}\beta$ for $SU(2)$ in $d=3$, $\approx 1.9\cdot 10^{19}\beta$ for $SU(3)$ in $d=4$, and $\approx 3.2\cdot 10^{14}\beta$ for $U(1)$ in $d=4$; the condition $\delta\le\frac{1}{2}e^{-48(d-1)\theta}$ holds for $\beta\le 4.8\cdot 10^{-17}$, $5.0\cdot 10^{-16}$, $2.6\cdot 10^{-20}$ and $1.5\cdot 10^{-15}$ respectively. These numbers have no physical meaning. They come from comparing the exact leading term, in which each of the four faces carries the dimension-reduced weight $u\approx(1+\iota)\beta/(2N^2)$, with sup-norm bounds on the remainder, in which each plaquette carries $\approx 4\beta$ times a polymer-counting factor $\approx eK$. A bound on the remainder through the character expansion, giving each face of a closed surface a weight of order u, would remove most of this loss; it is not attempted here. What the theorem establishes is the asymptotic statement of Corollary 5.5.

6 Two dimensions; discussion

Proposition 6.1 (Two dimensions)**.** *Let $d=2$ and $0<\beta<\beta_0(2)$, so that $K=4$ and $q(\beta)<0.909$ (Lemma 4.2).*

(i) For every $L_1\ge 3$, the spatial lattice $\Lambda_s=\mathbb{Z}/L_1\mathbb{Z}$ has $m_{\Lambda_s}(\beta)\ge L_1\log(1/q(\beta))$.

(ii) In infinite volume, $\mathrm{Cov}_{\mu_\beta}(A,B)=0$ for all A,B as in Theorem 1.2(iii) (time ranges with $a_2<b_1$); hence $m(\mu_\beta)=\infty$.

Proof. By Lemma 3.1, $g=L_1$, and Theorem 4.1 gives the bound with $q^{L_1\tau}$. (i) follows as in the proof of Theorem 1.2(i), with L_1 in place of 4. For (ii), the bound for the periodised observables on tori with $L_0,L_1\to\infty$ tends to 0, since $\tau\ge 1$ and $g=L_1\to\infty$, and the torus covariances converge to $\mathrm{Cov}_{\mu_\beta}(A,B)$ as in the proof of Theorem 1.2(iii) (note $\beta_0(2)<\beta_*(2)$, Lemma 4.2); then [II, Theorem `iv:thm:OS`](iii) applies with every $m>0$. □

Remark 6.2 (Consistency with the exact solution). In $d=2$ the transfer matrix on a spatial circle of length L_1 is diagonal in the characters of the holonomy, with eigenvalue ratios $(c_r/(d_rc_0))^{L_1}$ (`lattice/ymlattice/transfer2d.py`, tested against the Monte Carlo); so the exact finite-volume gap is $L_1\log(1/u(\beta))$ for $SU(2)$, linear in L_1, and the infinite-volume theory has no propagating states. Proposition 6.1 reproduces both features (with q in place of u). It is also consistent with [II, Proposition 'Gauge invariance of the ground-state chain' (`iv:prop:phys`)]: in $d=2$ the spatial links of a time slice form a line, gauge-invariant functions of finitely many of them are constant, and P acts trivially on the physical sector. The difference between $d=2$ and $d\ge 3$ is the girth of the spatial graph (L_1 versus 4): in $d\ge 3$ a closed surface can turn a corner in every time layer with four plaquettes, the minimal glueball.

Remark 6.3 (The strong-coupling test, sharpness, and the $U(1)$ test). (i) Principle 3 of the research program requires the correct asymptotics $m(\beta)\sim 4\log(1/u(\beta))$. Theorem 1.2 gives the lower half, $m\ge 4\log(1/\beta)-O(1)$ for all $\beta<\beta_0(d)$, and Theorem 5.4 the upper half,

$m \le 4\log(1/u(\beta)) + O(\beta)$ for sufficiently small β; Corollary 5.5 combines them. The upper bound uses a single correlator at time separations 0 and 1; by the positivity of P ([II, Remark 'Consequences for the numerics' (`iv:rem:numerics`)]) any larger pair of separations could be used instead. The additive constants of the two halves differ ($-4\log(4e^3K)$ versus $+4\log\frac{2N^2}{1+\iota}$); the strong-coupling expansion predicts $m = 4\log(1/u) + O(u)$, i.e. the upper one. (ii) The constant $-4\log(4e^3K)$ reflects the crude counting of [III] and the crude bound $\|g_p\|_\infty \le e^{4\beta} - 1$: for $SU(N)$ and $U(1)$, $c_0 = 1 + O(\beta^2)$ gives $\|g_p\|_\infty \le 2\beta + O(\beta^2)$, which improves the constant by $4\log 2$, and counting only surfaces without free edges, as in [III, Remark 'Constants, sharpness, improvements' (`al:rem:constants`)](ii), would improve it further. (iii) The hypothesis $L_i \ge 4$ in (i) is needed: if a spatial side equals 3, Theorem 4.1 holds with $g = 3$, and closed strips of three plaquettes per layer around that side (torelons, carrying the flux of a Polyakov loop) are expected (heuristic) to give states of energy $\approx 3\log(1/u(\beta))$, below $4\log(1/q(\beta))$ for very small β. (iv) Nothing here is non-abelian: the argument applies verbatim to $G = U(1)$, whose $d = 4$ theory is expected to be massless at large β (proved in [FS82] for the Villain action and asserted there for Wilson's; [III, Remark 'The $U(1)$ test' (`al:rem:u1test`)]). The result therefore concerns the strong-coupling endpoint only; it shows that the endpoint is controlled with the right leading behaviour of the lower bound, which is what a renormalisation-group argument would need to reach.

Summary of results

Statement	Where	Status
Duplicated system; loop expansion with Ξ^2; cancellation of non-connecting terms	Lem. 2.2, 2.4, 2.5	Theorem
Surfaces without free edges have $\ge g$ temporal plaquettes per crossed layer ($g = 4$ for $d \ge 3$, spatial sides ≥ 4)	Lem. 3.1, 3.2, 3.3, Cor. 3.4	Theorem
Covariance bound $q^{g\tau}$ in finite and infinite volume	Thm. 4.1, 1.2(iii)	Theorem
$m(\mu_\beta) \ge 4\log(1/q(\beta)) = 4\log(1/\beta) - O(1)$, and $m_{\Lambda_s}(\beta) \ge 4\log(1/q(\beta))$ for spatial sides ≥ 4; $d \ge 3$, $\beta < \beta_0(d)$	Thm. 1.2	Theorem
$d = 2$: gap $\ge L_1\log(1/q)$; infinite gap in infinite volume	Prop. 6.1	Theorem
Leading term of the plaquette–plaquette covariance: exactly $\tau_0 u^4$ (four sides of a cube); unique minimal connecting set	Lem. 5.1, 5.2	Theorem
Matching upper bound $m \le 4\log(1/u(\beta)) + O(\beta)$ (infinite volume; finite volume, sides ≥ 3), for sufficiently small β	Thm. 5.4	Theorem
$m(\mu_\beta) = 4\log(1/\beta) + O(1)$ as $\beta \downarrow 0$ (also finite volume, sides ≥ 4)	Cor. 5.5	Theorem
Upper bound for β of order 10^{-4} (character-expansion bound for the remainder)	Rem. 5.6	open

References

[F] *Lattice Yang–Mills theory: foundations for the mass gap problem*, Part I of the project "Solving Yang–Mills and the Mass Gap", `theory/lattice-foundations.tex` (this repository), living document.

[II] *Lattice Yang–Mills theory at strong coupling II: the infinite-volume theory, its transfer operator and its mass gap*, `theory/infinite-volume-strong-coupling.tex` (this repository), living document, version of 9 November 2026.

[III] *Lattice Yang–Mills theory at strong coupling III: confinement (area law)*, `theory/area-law-strong-coupling.tex` (this repository), living document, revised 9 November 2026.

[FS82] J. Fröhlich, T. Spencer, Massless phases and symmetry restoration in abelian gauge theories and spin systems, Commun. Math. Phys. 83 (1982), 411–454. (verify)

[OB85] M. O'Carroll, W. D. Barbosa, J. Math. Phys. 26 (1985): glueball masses in $3+1$ dimensional lattice gauge theory at strong coupling (title and content to be checked; only an abstract has been seen). (verify)

[OBS85] M. O'Carroll, G. Braga, R. Schor, Commun. Math. Phys. 97 (1985): glueball masses in $2+1$ dimensional lattice gauge theory at strong coupling (title and content to be checked; only an abstract has been seen). (verify)

[Sch83] R. Schor, Existence of glueballs in strongly coupled lattice gauge theories, Nucl. Phys. B222 (1983), 71–82. (verify)